

Research Statement

My research focuses on the Economics of Education particularly for disadvantaged students in developing countries. I design and test practical solutions to problems that bottleneck learning, or keep students out of school. I also develop theory to explain how students form foundational reading skills crucial to becoming full participants in a modern economy.

Learning is a complex beast that depends on the conflicting interests of many stakeholders. In my job market paper, I cross-randomize two interventions that target parents and administrators. Understanding how these stakeholders interact is key to building scalable literacy programs. Parents, teachers, and/or administrators adjusting their effort in response to a literacy program is a key, poorly understood, reason why seemingly successful programs see their effects drop off when brought to hundreds or thousands of schools. Being able to scale successful programs is crucial to solving the learning crisis. My job market paper furthers this research agenda. I cross-randomize two interventions: 1) organized parent meetings that give parents tools to understand how literate their child is, and encourages reading at home 2) structured management training of principals on how to coach their teachers to more effectively teach literacy and interact with parents. The key mechanism through which each program works is through unexpected interactions with inputs to the interventions. In particular, the management training for principals decreased reading ability for treated students because the additional coaching pushed teachers to improvise during their lessons more which reduced their ability to adapt to the needs of their students. By cross-randomizing the interventions, I further study the issue of stakeholder interactions by seeing how parents, teachers, and students respond when they receive *both* interventions. Parents (or siblings) increase their effort supporting students at home, and students practice reading more frequently. This increase in effort was enough to almost fully offset the decrease in reading ability from the principal intervention alone. This is a key mechanism for why some education programs succeed and why some fail particularly at scale.

I am particularly interested in studying exactly how learning occurs. This goes beyond simple impact evaluations of education programs asking if the average student improved to uncover who the important stakeholders are in producing learning (students, parents, teachers, administrators) and what mechanisms drive successful programs. I do this by combining randomized field experiments with cutting-edge statistical and econometric techniques and economic theory. For example, in my study "How to Build a Reader: Evidence From a Scalable Literacy Intervention in Ghana, my co-authors and I tested a large scale education program that combined structured pedagogy and targeted instruction. We found the program increased literacy among first grade students in rural Ghana by a huge amount over the control group. Using economic theory, we were able to learn even more from the data about the mechanisms of the program: I developed a model of interlocking skill formation and showed it is consistent with the data. The model let us understand which students were most impacted by the program, and how the program could best target instruction at specific literacy skills and in what order.

In a related line of work, I am working with a separate team of co-authors on a project in North-West Nigeria to study the effects a program designed to prevent gender-based violence in schools. This intervention is also aimed at alleviating impediments to education through a different channel. Our preliminary research shows that the program raises awareness to this sensitive issue among teachers and students, decreases the prevalence of gender-based violence, and may raise literacy levels. We are currently applying for grants with the aim to expand the program to 300 schools next school year.

My research is not confined to Africa nor to elementary school students. I have an ongoing study at a medium-sized university in California working to understand how to mitigate the negative effects of academic probation. The particular challenge in this setting is the university lack resources to allocate to proven methods to help guide students to academic recovery such as heavy-touch counseling sessions, so we redesigned the administrative letter

sent to students when they are put on probation using softer language and highlighting concrete strategies they can use to get off probation. We are in the process of tracking a cohort of students who received our modified letter over the course of multiple years to see if they drop out less than their peers. This research studies administrative barriers to learning faced by students.